



440C Martensitic Stainless Steel (UNS S44004)

Categories: [Metal](#); [Ferrous Metal](#); [Martensitic](#); [Stainless Steel](#); [T 400 Series Stainless Steel](#)

Material Fe-17Cr-1Mn-1Si-0.75Mo-1.1C

Notes: Typically used for molds, bearings, turbine components, aircraft structural parts, medical instruments, and cutlery

Key Words: UNS S44004

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

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Physical Properties	Metric	English	Comments
Density	7.80 g/cc	0.282 lb/in ³	
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell B	97	97	
Hardness, Rockwell C 	50	50	
	@Temperature 416 °C	@Temperature 780 °F	
	52	52	
	@Temperature 316 °C	@Temperature 600 °F	
	56	56	
	@Temperature 204 °C	@Temperature 400 °F	
	60.5	60.5	
	@Temperature 93.3 °C	@Temperature 200 °F	
	63	63	
	@Temperature -17.8 °C	@Temperature 0.000 °F	
Hardness, Vickers 	510	510	
	@Temperature 416 °C	@Temperature 780 °F	
	555	555	
	@Temperature 316 °C	@Temperature 600 °F	
	630	630	
	@Temperature 204 °C	@Temperature 400 °F	
	710	710	
	@Temperature 93.3 °C	@Temperature 200 °F	
	760	760	
	@Temperature -17.8 °C	@Temperature 0.000 °F	
Tensile Strength, Ultimate	760 - 1970 MPa	110000 - 286000 psi	
Tensile Strength, Yield	450 - 1900 MPa	65300 - 276000 psi	
	@Strain 0.2 %	@Strain 0.2 %	
Elongation at Break	2.0 - 14 %	2.0 - 14 %	

Modulus of Elasticity	204 - 215 GPa	29600 - 31200 ksi	
Bulk Modulus	166 GPa	24100 ksi	
Poissons Ratio	0.283	0.283	
Fracture Toughness	17.6 - 24.2 MPa-m ^{1/2}	16.0 - 22.0 ksi-in ^{1/2}	cracks originating in case; case depth 0.30 to 0.35 in.
Shear Modulus	83.9 GPa	12200 ksi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000600 ohm-cm	0.0000600 ohm-cm	

Thermal Properties	Metric	English	Comments
CTE, linear	10.2 µm/m-°C	5.67 µin/in-°F	
Specific Heat Capacity	0.460 J/g-°C	0.110 BTU/lb-°F	
Thermal Conductivity	24.2 W/m-K	168 BTU-in/hr-ft ² -°F	

Component Elements Properties	Metric	English	Comments
Carbon, C	1.1 %	1.1 %	
Chromium, Cr	17 %	17 %	
Iron, Fe	79.15 %	79.15 %	As Balance
Manganese, Mn	1.0 %	1.0 %	
Molybdenum, Mo	0.75 %	0.75 %	
Silicon, Si	1.0 %	1.0 %	

[References](#) for this datasheet.

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